Denver SPEED Program

2024 Improvements & 2025 Action Plan

Summer 2025

DENVER THE MILE HIGH CITY × VISIONZERO

SAFER STREETS SAVE LIVES

SPEED Program Agenda

1. Program Overview & Vision Zero Goals
2. Equity Analysis
3. Safety Assessment
4. Current & Future Program Improvements
5. Community Engagement Plans
6. Next Steps

SPEED Program Overview

Denver's Strategic Transportation Plan – Denver Moves Everyone

- Building a world class transportation system by 2050
- Moving everyone and everything equitably, safely, and sustainably

Denver's Vision Zero Goals

- Zero traffic-related death or serious bodily injury by 2030
- Reduce traffic fatalities by 50% on Federal and Alameda by 2026

Five Elements of the Safe Systems Approach

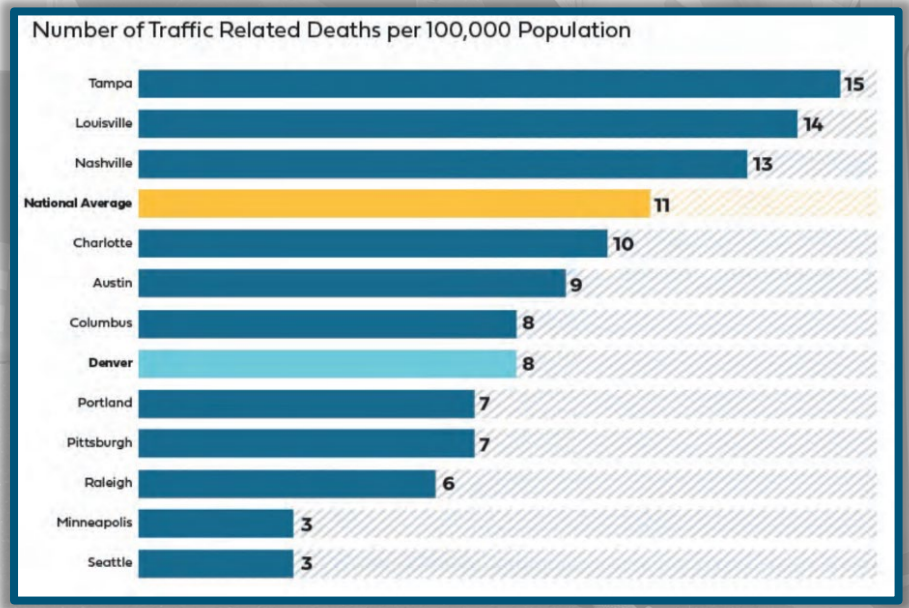
- Street systems are planned, designed, and operated to:
  - anticipate inevitable human mistakes
  - mitigate the severity of crashes
  - decrease the likelihood of severe injury or death
- Every journey is inherently safe for all through:
  - **Safe streets, safe speeds, safe road users,** safe vehicles, & post-crash care

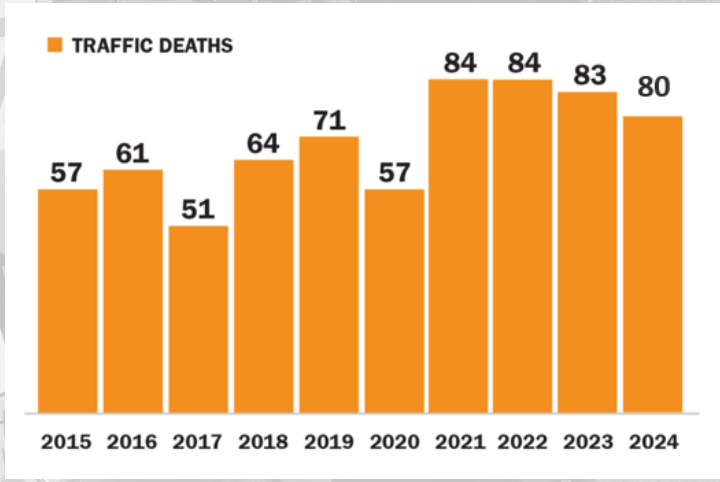
PROMOTING SAFER SPEEDS

DENVER'S NEW SPEED PILOT PROGRAM IS ADDRESSING SPEEDING ON HIGH-RISK STREETS, WITH A FOCUS ON FEDERAL BLVD AND ALAMEDA AVE

- ✓ Saturate corridors with speed limit signs and speed feedback signs to provide visual reminders of posted limits
- ✓ Program traffic signals to slow vehicles down to the posted limits
- ✓ Engineer modified signal timing to prevent excessive speeding at night
- ✓ Expand use of signal timing that gives pedestrians a head start crossing the street before the light turns green
- ✓ Deploy speed enforcement including mobile speed vans and piloting automated speed enforcement cameras

Vision Zero Crash Data

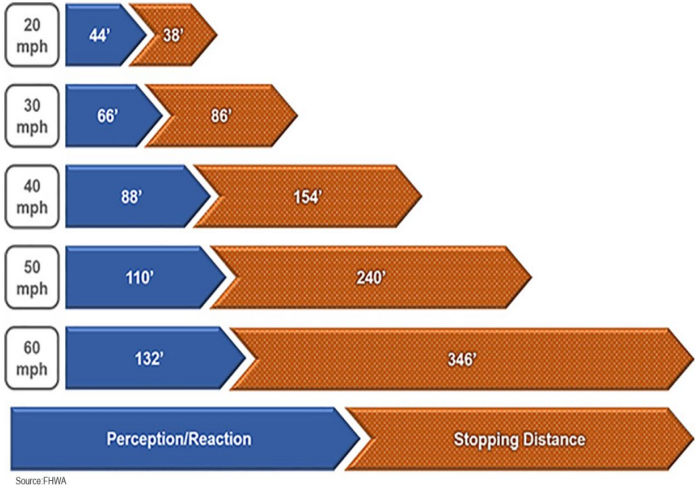


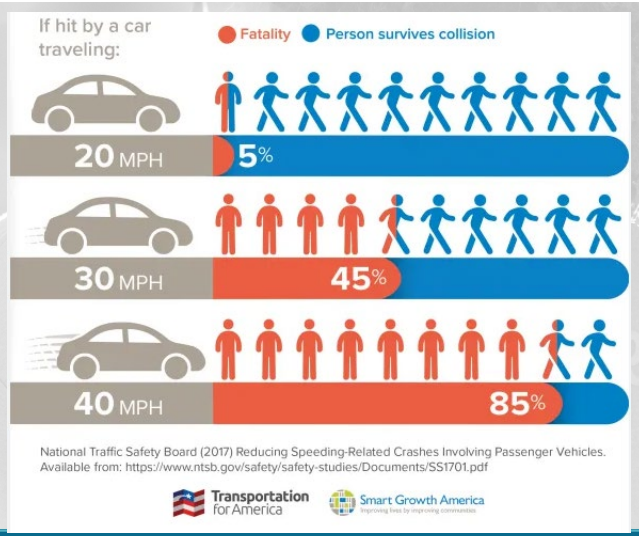


Speed is 1 of the 5 Elements That Create the Safe System

Safe Speeds & Automated Enforcement

High-speed crashes are deadly. Reducing speeds can improve safety outcomes by reducing impact forces, providing additional time for drivers to stop, and improving visibility.





DPD Automated Enforcement Outcomes

6th & Lincoln Red Light Camera

- Total crashes decreased 52%
- Side impact crashes decreased 75%
- Injury crashes decreased 80%

DPD Mobile Units

- 5-day deployment sees an average 21% reduction in excessive speeding

Speeding- Related Crash Analysis

Dataset for speeding, or driving recklessly, or driving carelessly at time of crash.
Crash data from Jan 2020 to May 2024 obtained from Denver Police Department (DPD).

Facility Name	Total Crashes	Speeding Crashes	Reckless Crashes	Careless Crashes
Colfax Avenue	3,915	57	36	376
Federal Boulevard	3,567	64	31	357
Colorado Boulevard	3,376	54	19	313
6th Avenue	2,582	48	20	259
Alameda Avenue	2,566	48	15	236
Quebec Street	2,266	58	15	200
Broadway	2,196	38	18	161
Sheridan Boulevard	2,078	49	18	208
Speer Boulevard	1,922	36	9	131
Evans Avenue	1,919	53	17	192

Equity Analysis

Equity & Harm Reduction Priorities

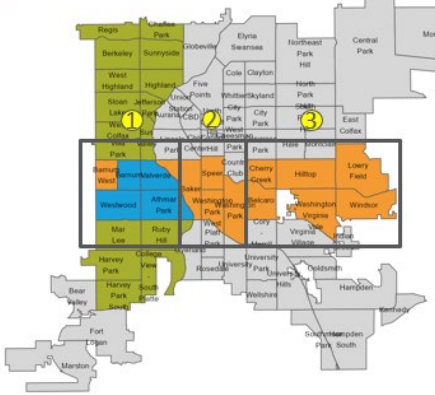
- OSEI partnership, leadership, and advisement on **equity framework** and **evaluation**
- Interagency MAPS process *(3 risk categories)*
- Sharp focus on **data collection**, **performance metrics**, and **outcomes** to:
  - **Achieve** safety goals and positive behavior changes
  - **Invest** in safety infrastructure in marginalized communities
- 30 Harm Mitigation Strategies recommended to:
  - **Address** disproportionate impacts on vulnerable communities *(risk 1)*
  - **Prevent** further degradation of trust *(risk 2)*
  - **Mitigate** legal or financial burden on adjacent communities *(risk 3)*

GOALS

- ✓ Safety
- ✓ Equity
- ✓ Transparency
- ✓ Mobility
- ✓ Community
- ✓ Quality of Life

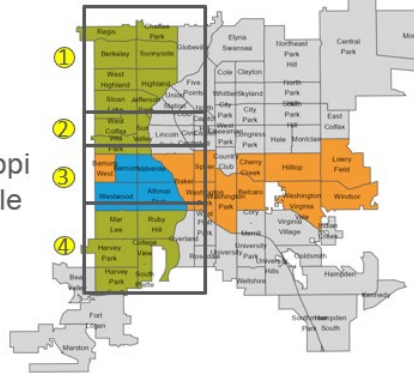
Neighborhoods along Alameda Ave. were separated into three segments.

1. Pecos to Sheridan
2. Lincoln to University
3. Colorado to Quebec



Neighborhoods along Federal Blvd. were separated into four segments.

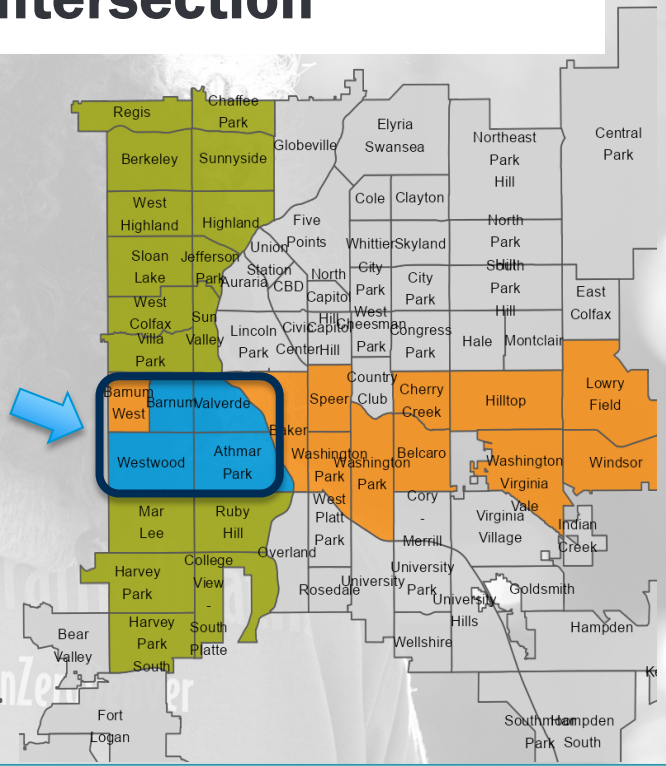
1. I-70 to Colfax
2. Colfax to US 6
3. US 6 to Mississippi
4. Mississippi to Yale



DATA ANALYSIS: Federal & Alameda

EQUITY ANALYSIS: Alameda & Federal Intersection

- All the neighborhoods have **between 1.5x and 3x more people living in poverty** than the Denver average (12%).
- All the neighborhoods have a **lower median household income** than the Denver average ($81,916).
- All the neighborhoods are **more racially and ethnically diverse** than Denver.
- All the neighborhoods have a **larger population of people under the age of 30** than Denver's overall population (39%).
- All the neighborhoods **have a higher proportion of people with a High School diploma or less** compared to Denver (28%).
- In Athmar and Valverde, at least 50% of people **commute 30 minutes or less**.
- In Barnum and Westwood, at least 50% of people **commute 30 minutes or more.**



Harm Mitigation Measures – Avoid Disproportionate Impacts

- ☑ Establish transparent and equitable Installation Criteria.
- ☑ Implement and evaluate safety countermeasures.
- ☑ Conduct additional equity analysis prior to city-wide implementation.
- ☑ Evaluate corridors prior to camera install.
- ☑ Clear establishment of thresholds to focus on excessive speeders.
- ☑ Clear signage location and maintenance requirements.
- ☑ Consider installing additional signage/graphics beyond legal requirements and design standards.
- ☑ Develop review & evaluation requirements including frequency of assessing equity and technical data.
- ☑ Investigate alternatives to fees and fines to mitigate potential financial burden.

Harm Mitigation Measures – Prevent Degradation of Trust

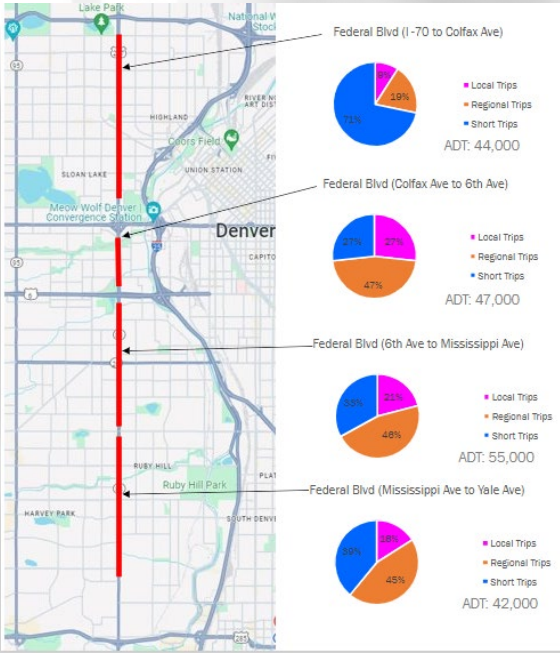
- ☑ Conduct Aggressive Engagement and Transparency Campaign.
- ☑ Publicly announce the implementation of the system online and along designated corridors.
- ☑ DOTI to be the lead and public face of the program, including issuance of citations.
- ☑ Publication of program metrics in an easily accessible and consistent manner.
- ☑ Investigate using revenue to fund safety projects, programs, and infrastructure.
- ☑ Consider not capturing driver's faces unnecessarily and destroying all personal identifiable information.
- ☑ Contract with Academic Partner or Independent Evaluator to set metrics, criteria, and evaluation.
- ☑ Feedback signs should be installed without blue & red flashing lights.
- ☑ Placement of cameras should consider areas with the highest risk of disproportionate impact.

Harm Mitigation Measures – Mitigate Financial Burdens

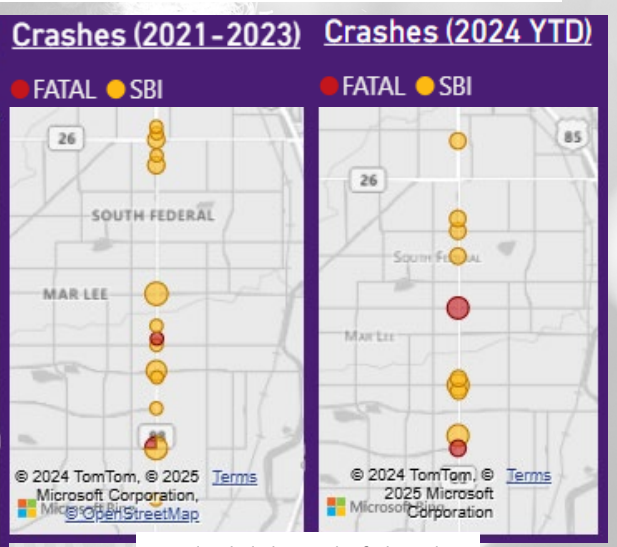
- ☑ Consider extending grace period beyond 30 days and evaluation of grace period data prior to enforcement.
- ☑ Implement equitable fine structure and policies.
- ☑ Deadline for payment should meet or exceed state statute.
- ☑ Reduce number of days between the violation and citation issuance
- ☑ Clear mechanisms and procedures for contesting violations.
- ☑ Educational materials and official documents must be available in multiple languages and included with citation.
- ☑ Only allowable penalty should be the initially assessed fine or fine alternative.
- ☑ How to handle violations without payment.
- ☑ Non-financial penalties restricting vehicle access and use shall not be allowed.
- ☑ Vender shall not dictate fee structure or camera placement.
- ☑ Clear procedures and policies established for evidence review and due process.

Safety Assessment

SAFETY ANALYSIS: Federal Vehicle Trips & Crash Data



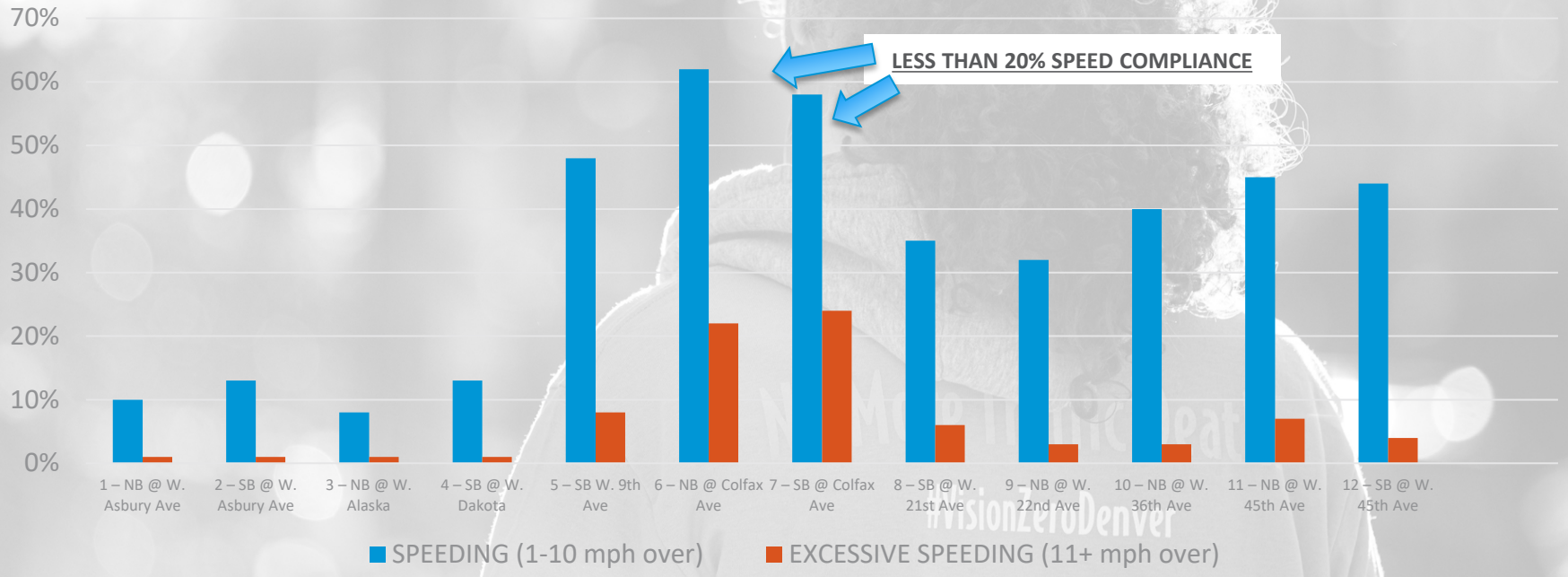
Federal Blvd – North of Alameda



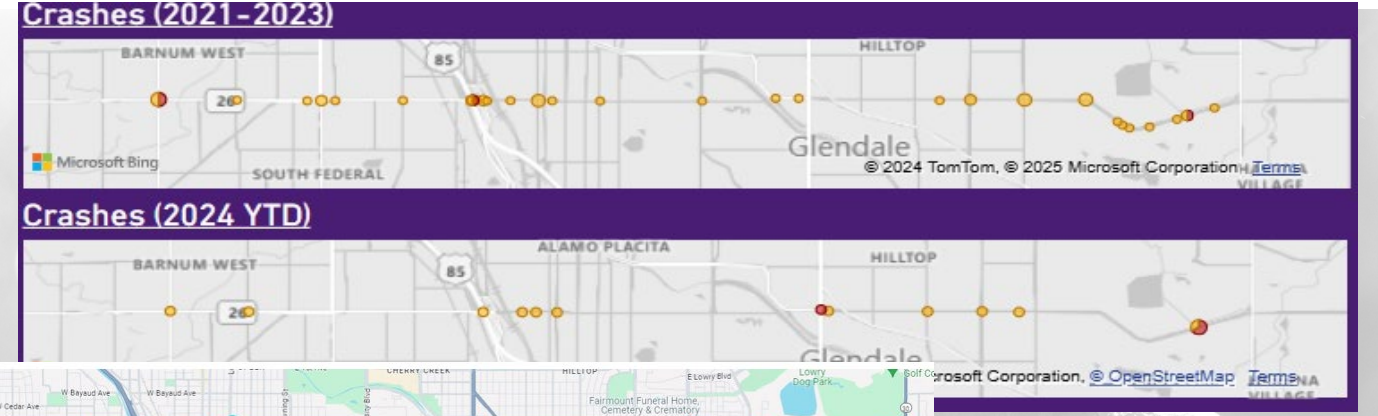
Federal Blvd – South of Alameda

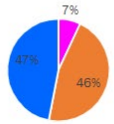
SAFETY ANALYSIS: Federal Blvd Speed Data

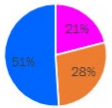
CDOT installed twelve TDC2 units along Federal Blvd, collecting speed data 24 hours/day for six (6) days.

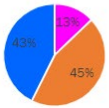


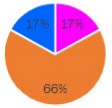
SAFETY ANALYSIS: Alameda Vehicle Trips & Crash Data











- Local Trips
- Regional Trips
- Short Trips

Key:

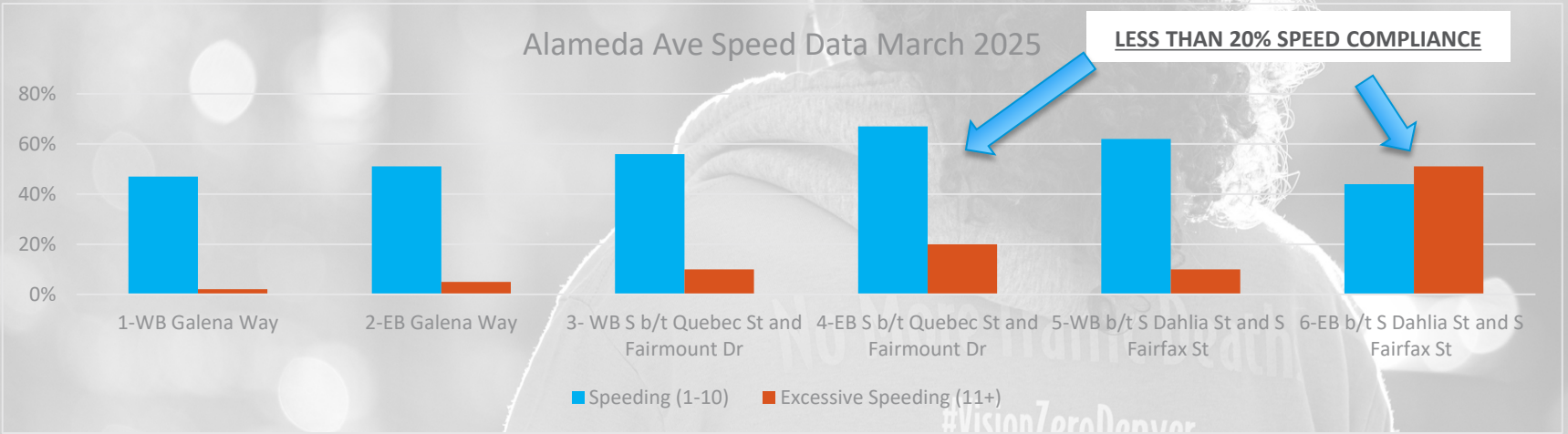
Local trips: Trips by people who live or work along Alameda Ave and drive a much longer distance on Alameda Ave. These trips are likely to be impacted by speed enforcement.

Regional trips: Regional trips that use Alameda Ave. These trips are likely to be impacted by speed enforcement.

Short trips: These trips use Alameda Ave for a very short distance (2-3 blocks). These trips are not likely to be impacted by speed enforcement.

SAFETY ANALYSIS: Alameda Speed Data

DOTI contracted TDC2 units along E Alameda Ave, collecting speed data 24 hours/day for one (1) day.



SPEED Program Improvements

SPEED Program Overview

S

Saturating corridors with speed limit and speed feedback signs, providing visual reminders of posted limits and actual speeds.
Start directed marketing program to encourage drivers to slow down.

Completed

- ☑ Installed **23 electronic speed feedback signs** on Alameda Ave (9) and on Federal Blvd (14).
- ☑ Installed approximately **one speed limit sign every ¼** mile along Alameda Ave and Federal Blvd.
- ☑ CDOT conducted **speed study** of Federal Blvd.
- ☑ **"For the Love of Safety"** campaign on billboards and online to spread awareness of the danger of speeding and its relationship to traffic fatalities.

In Progress

- ❑ Facilitate grass roots **focus groups** to hear directly from the communities impacted.
- ❑ Distribute **SPEED program flyer** in "Wastewise" newsletter sent to 180,000 Denver households.
- ❑ **Safety is a Team Sport** Broncos Yearbook Ads

Communications & Community Engagement

- **Concept of operations plan** outlining the SPEED program's purpose, goals, scope, operations, and performance metrics.
- **Educate**, **inform**, **consult**, and **collaborate** with:
  - Community leaders, individuals, road users, adjacent neighborhoods, impacted communities, and the public.
- **Harm mitigation strategies** will **guide inclusive public engagement** consisting of:
  - Program website and resource library
  - Robust marketing campaigns
  - Focus groups and targeted business outreach
  - Press Releases, Mailers, and Newsletters
  - Language access and translation
  - City and partner social media and newsletters

✓ *Inclusive, trusted, and valued partner*

✓ *Education & Resources*

✓ *Collaboration & Feedback*

Communications & Community Engagement

Engagement as of July

- **4,067** total – 3,149 directly across 29 events; 708 indirectly through flyers/newsletters; 210 pledges
- **Upcoming**: DOTI Wastewater half-page mailer in July (179k), Broncos Player Yearbook in August (400k) with Nuggets, Avalanche, Rockies up next

Outreach materials developed

- Trifold, One-pager (Spanish & Vietnamese), Streamlined FAQ
- **Upcoming**: Social media posts, website updates, media toolkit

Focus groups

- Phase One focus group (44 people) held 4 meetings through June/July
- Top themes:
  - Speeding is a problem in our communities
  - Concerns and risks about surveillance and privacy
  - Reinvest in community, education, & traffic safety
  - Concerns about punitive actions when streets are designed for speed
  - Interest in data and impact of other safety countermeasures

SPEED Program Overview

P

Programming traffic signals to slow vehicles down to the posted limits.
(if you go the speed limit, you get more green lights)

- ☑ Completed **signal timing** on Federal Blvd
- ☑ Completed **signal timing** along Alameda Avenue

E

Engineering signal timing to prevent excessive speeding at night by eliminating Rest in Green.
(57% of fatal crashes last year happened at night)

- ☑ Completed **late night signal timing** on Federal Blvd
- ☑ Completed **late night signal timing** along Alameda Avenue

SPEED Program Overview

E

Expanding use of signal timing to give pedestrians a head start crossing the street before the light turns green.

Completed

- ☑ Completed **leading pedestrian** signal timing
- ☑ Refreshed crosswalks and pavement markings and installed daylighting at **Federal and 10th Avenue**

In progress

- ❑ Refresh crosswalks and pavement markings and installed daylighting at **additional priority intersections**

D

Deploying speed enforcement on both corridors including surging mobile speed vans and piloting automated speed enforcement cameras.

- ☑ Completed **equity analysis** of speed safety cameras and developed mitigations to unintended outcomes.
- ❑ Install **speed safety cameras** along Federal Blvd and Alameda Ave

Speed Safety Camera Considerations

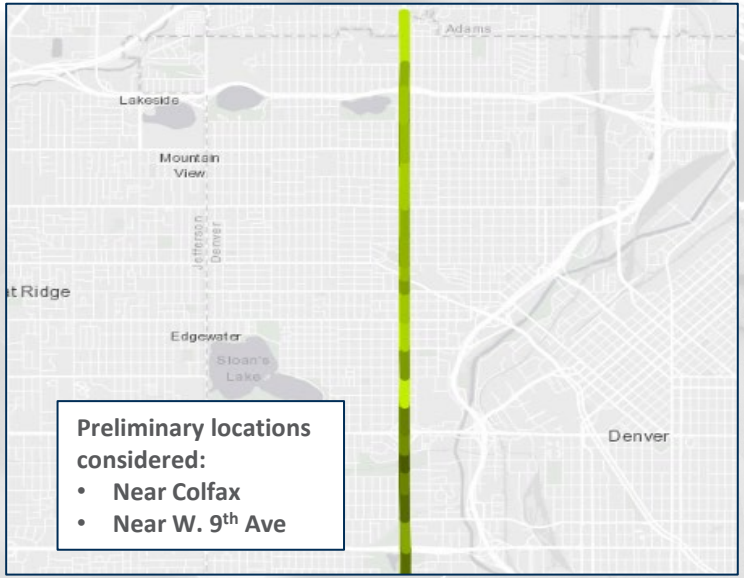
Location Considerations

- Areas of fatalities & serious bodily injuries (HINs)
- Areas of greatest speed incidences
- Areas with high pedestrian, bicycle, transit activity
- Complementary to DPD Mobile Units
- Schools and Parks
- Equity & community considerations
- Neighborhood Characteristics
- Topography & roadway geometry
- Planned/active capital improvement projects
- Ongoing behavior changes

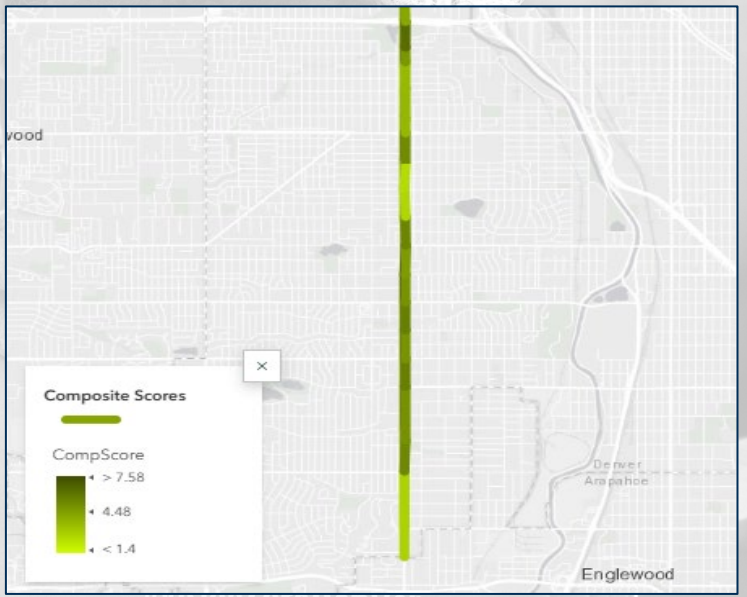
Fine Schedule

- Align with State and DPD fine structure
  - $40 base citation
  - $80 in school or construction zone

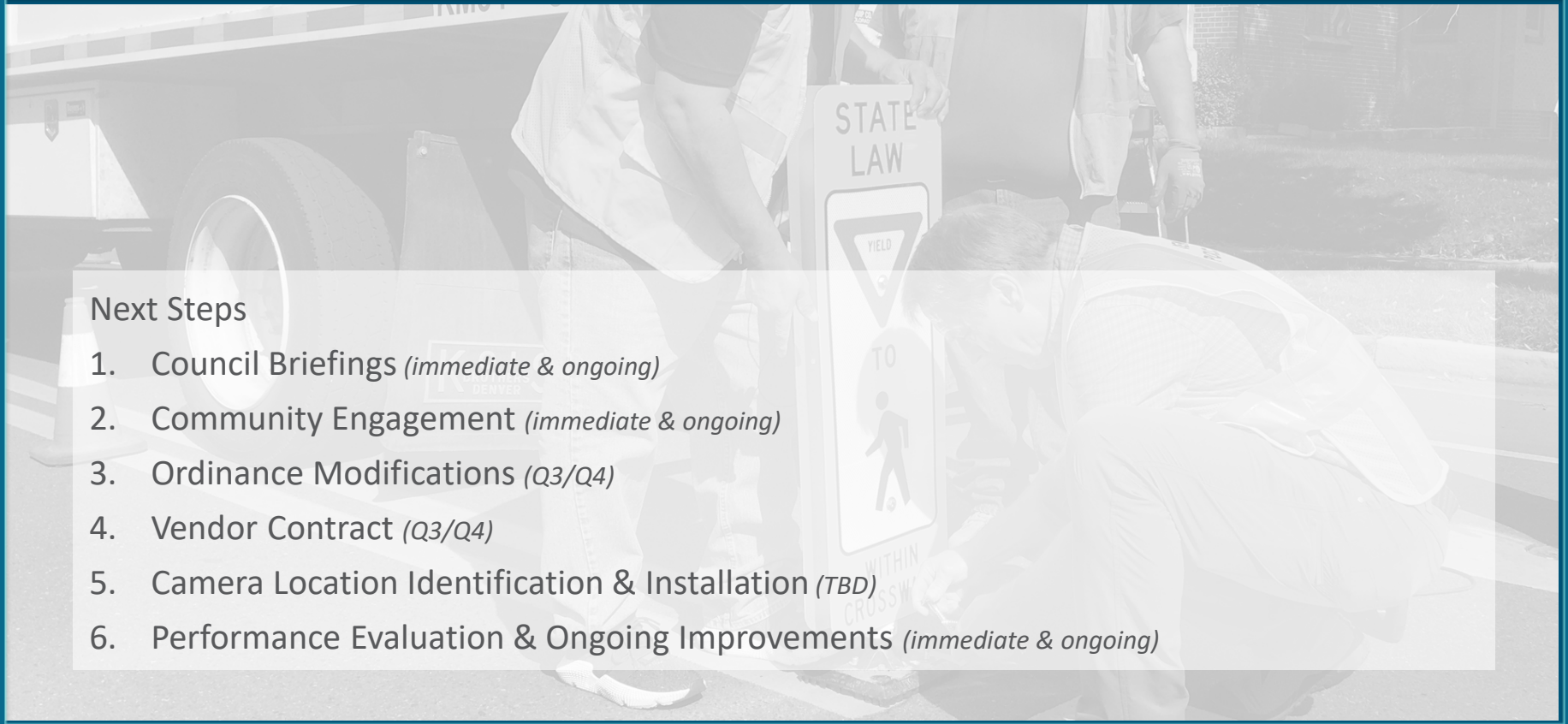
FEDERAL BLVD LOCATION CONSIDERATIONS



Federal Blvd – North of 6th Ave



Federal Blvd – South of 6th Ave



Next Steps

1. Council Briefings *(immediate & ongoing)*
2. Community Engagement *(immediate & ongoing)*
3. Ordinance Modifications *(Q3/Q4)*
4. Vendor Contract *(Q3/Q4)*
5. Camera Location Identification & Installation *(TBD)*
6. Performance Evaluation & Ongoing Improvements *(immediate & ongoing)*